Large-scale Single Tree Information Extraction of Oil Palm in Malaysia Based on Sub-meter Visible Light Satellite Images

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Abstract: The distribution of specific tree species and single tree information can shift down the research scale of geography and ecology, bridging research from regional to local scales. However, efficiently and finely extracting the spatial distribution and single tree information for a tree species regionally has always been a challenge due to the constraints of data and algorithms. With sub-meter visible light satellite images (RGB bands) in 2021 to 2022, we constructed a novel method for extracting the spatial distribution and single tree information of oil palm across Malaysia, namely the Single Tree Extraction Method - Oil Palm (STEM-OP). We developed a graphic morphological algorithm based on gradient difference window detection to extract the spatial distribution of oil palms. Then, we proposed an algorithm of double ring sliding detection algorithm to identify the canopy texture features of single oil palm. Finally, an adaptive iterative erosion algorithm was employed to separate overlapping and adjacent canopies into single trees. The overall accuracy of both the spatial distribution and number of single oil palm extracted by STEM-OP exceeded 90%. Our study revealed that the total area of oil palm in Malaysia was 5.25 million hectares, with a total of 572.6 million trees and an average planting density of 109 palms per hectare. The STEM-OP addresses the computational inefficiencies of methods with training samples and limitations of satellite image bands. For the first time, our study extracts the refined spatial distribution of oil palm and obtains the planting density and single tree position on a national scale. All of these are of great value to the formulation of oil palm development policies and ecological protection measures.

Keywords: RGB image; texture feature; morphology; double ring sliding detection; single tree

1 Introduction

Palm oil is consumed directly by more than 3 billion people worldwide and has significant applications in non-food industry (Carter et al., 2007; Ilyas et al., 2022; Rizeei et al., 2018; Srestasathiern and Rakwatin, 2014; Zhao et al., 2024). By providing accurate data on the spatial distribution of oil palm plantations and the number as well as position information of single oil palm, we can offer multiple stakeholders reliable information, thereby promoting the sustainable development of the oil palm industry (Xu et al., 2022; Yin et al., 2023). Additionally, extracting single oil palm information helps bridge the gap between regional and local scales, which is crucial for optimizing planting management and ecological protection (Guillaume et al., 2018; Koh et al., 2011; Miettinen et al., 2016).

The tree species and the single tree constitute the fundamental component of forests (Li et al., 2018; Li et al., 2016). Forest tree species classification and single tree information extraction are vital for forest resource inventories, biodiversity monitoring, biomass and carbon stock accounting, and the formulation of sustainable management and protection strategies (Fang et al., 2020; Fassnacht et al., 2016; Sun et al., 2019). Currently, tree species extraction methods based on high-resolution satellite images have made significant progress, including pixel-based classification, object-oriented classification, and spectral mixture analysis classification (Goldblatt et al., 2018; Li et al., 2019a; Liu and Wu, 2018; Ma et al., 2017; Pham et al., 2016; Pu, 2021; Zhang et al., 2020b). Additionally, deep learning methods in combination with remote sensing data have been widely applied (Mayra et al., 2021; Zhang et al., 2020a; Zhang et al., 2021). For example, Pouliot et al. (2002) proposed a canopy positioning and extraction method that achieved an accuracy of 93% in a study area of approximately 0.017 km². Shafri et al. (2011) used a single tree detection method based on shadow extraction and template matching to successfully detect 75% of trees in a study area of about 95 km². Malek et al. (2014) proposed a single tree detection method based on SIFT operator feature extraction and an extreme learning machine classifier, achieving an identification accuracy of over 89% in a study area of about 0.057 km². Additionally, with the advent of deep learning, single tree detection methods continue to evolve (Zheng and Wu, 2022). Wang et al. (2019) converted LiDAR data into images through front and side projection transformation and then input the images into a deep learning network, achieving an overall accuracy of 90% in rubber tree extraction in an experimental plot of 1.08 km². Weinstein et al. (2019) proposed a semi-supervised method using unsupervised algorithms and point cloud data to generate labeled data. Osco et al. (2021) used RGB images from airborne observation platforms to detect the number of trees in high-density plantations through deep learning networks, achieving an overall accuracy of over 90%. However, both machine learning and deep learning methods require extensive sample selection, learning, model training, and classification operations (Arevalo-Ramirez et al., 2021; Dechesne et al., 2017), making it impractical for current studies to map more than a million trees over a large area (Zheng et al., 2023). Even in the extraction of millions of trees, forest areas with high canopy density are often considered as closed tree canopies, and single trees are hard to be separated (Brandt et al., 2020; Reiner et al., 2023).

With regard to the oil palm in particular, there is an increasing amount of research into its extraction. Jebur et al. (2014) achieved overall accuracies of 81.25% and 76.67% using object classification and pixel analysis, respectively. Xu et al. (2020) combined MODIS images and synthetic aperture radar data to produce annual oil palm plantation data for four major oil palm-producing countries in the Asia-Pacific region (Indonesia, Malaysia, Thailand, and Papua New Guinea) from 2007 to 2018. Descal et al. (2021) employed deep learning methods to create a 2019 global oil palm plantation map, distinguishing between industrial and smallholder plantations. Dong et al. (2020) developed a new segmentation method to map oil palm land use in complex environments. In terms of single tree information extraction of oil palm, Li et al. (2016a) utilized convolutional neural networks and sliding window technology to detect oil palm from QuickBird high-resolution satellite images. Mubin et al. (2019) proposed a

CNN method combined with GIS processing and high-resolution Worldview-3 satellite images to capture object images and detect young and mature oil palm. Furthermore, Bonet et al. (2020) used the transfer learning CNN method and VGG-16 architecture for feature extraction to detect oil palm from drone images. Overall, previous studies have focused on extracting the spatial distribution of oil palms at different scales, primarily using multi-band data containing near-infrared spectral and LiDAR information. However, due to the limitations such as the resource data, training samples and computing resources required by these algorithms, few attempts have been made to extract single oil palm, and large-scale single oil palm data are greater lacking (Li et al., 2019b; Zheng et al., 2019).

In this study, given that Malaysia is one of the most intense and geographically concentrated oil palm planting area worldwide (Cheng et al., 2019; Xu et al., 2020), we thoroughly explored the morphological features of oil palms on satellite images, and constructed an oil palm remote sensing extraction method named the Single Tree Extraction Method - Oil Palm (STEM-OP) by using sub-meter RGB satellite images. In the process, the spatial distribution of oil palms was firstly extracted using a gradient difference window detection algorithm. Subsequently, we developed double ring sliding detection and adaptive iterative erosion algorithms to extract single tree information, including density and the positions of oil palm. The spatial distribution and single tree data obtained in this study provide new perspectives and insights for understanding the national-scale distribution of oil palms and formulating appropriate ecological protection strategies in Malaysia.

2 Study area and study data

2.1 Study area and accuracy assessment area

Malaysia is located between 1° - 7°N and 97° - 120°E in Southeast Asia. The country is divided into two parts by the South China Sea: West Malaysia (Peninsular Malaysia) and East Malaysia (on the island of Borneo). Malaysia has a total land area of about 330,000 km², with West Malaysia covering 132,000 km² and East Malaysia covering 198,000 km² (Fig. 1). Malaysia is divided into 13 states: Johor, Kedah, Kelantan, Malacca, Negeri Sembilan, Pahang, Perak, Perlis, Penang, Selangor (including the capital Kuala Lumpur and the federal territory of Putrajaya), Terengganu, Sabah (including the federal territory of Labuan), and Sarawak. As one of the world's major palm oil producers, Malaysia keeps a rapid increase in the area of oil palm over the past 30 years (Descals et al., 2021; Du et al., 2022; Tang and Al Qahtani, 2020).

To evaluate the accuracy of spatial distribution and single tree information of oil palm extracted in this study, we selected four pairs of latitude and longitude coordinates within the oil palm areas in each of the 13 states (Xu et al., 2020). These coordinates served as the center points for buffering to create a 1 km × 1 km grid, and generated a total of 52 accuracy assessment areas.

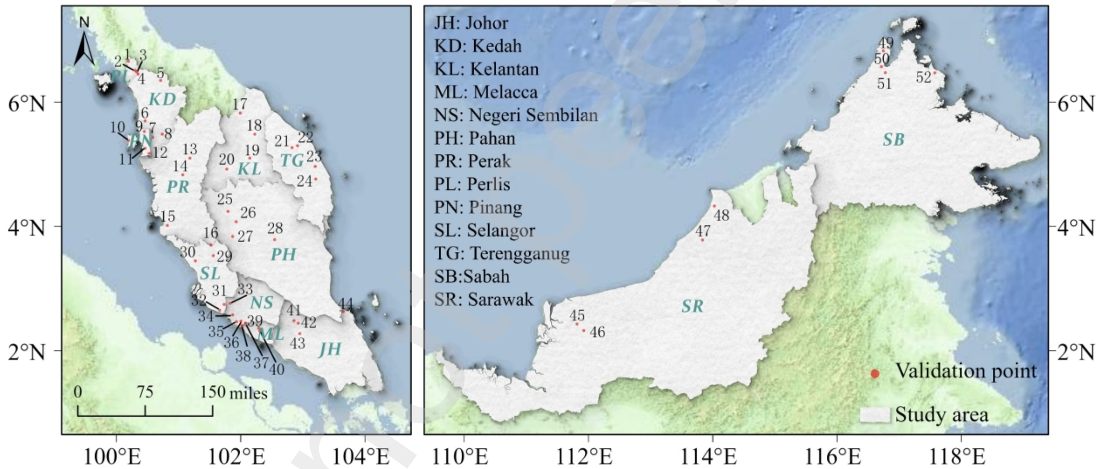

Fig. 1. Study area and the 52 accuracy assessment areas.

2.2 Study data and data preprocessing

2.2.1 Sub-meter high resolution satellite image

This study collected sub-meter high resolution satellite images from Google Earth in the years of 2021 and 2022 (Zhang et al., 2023), including QuickBird-2, GeoEye-1, WorldView-2, and WorldView-3 sub-meter visible light RGB satellite images. The study area was divided into 42,360 image units with each covering an area of 7.96 km². The dimensions are approximately 3.5 × 2.3 km (1.0 GB) with a spatial resolution of ≈ 0.15 m. Excluding 1,889 images with no available data and 1,186 images with significant cloud cover, a total of 39,285 sub-meter visible light satellite images were collected, amounting to 39.90 TB and accounting for 92.74% of all image units in the Malaysian land area. Among these, 7,722 image units included image stitching situations come from various acquisition time or various sensor images (Fig. 2).

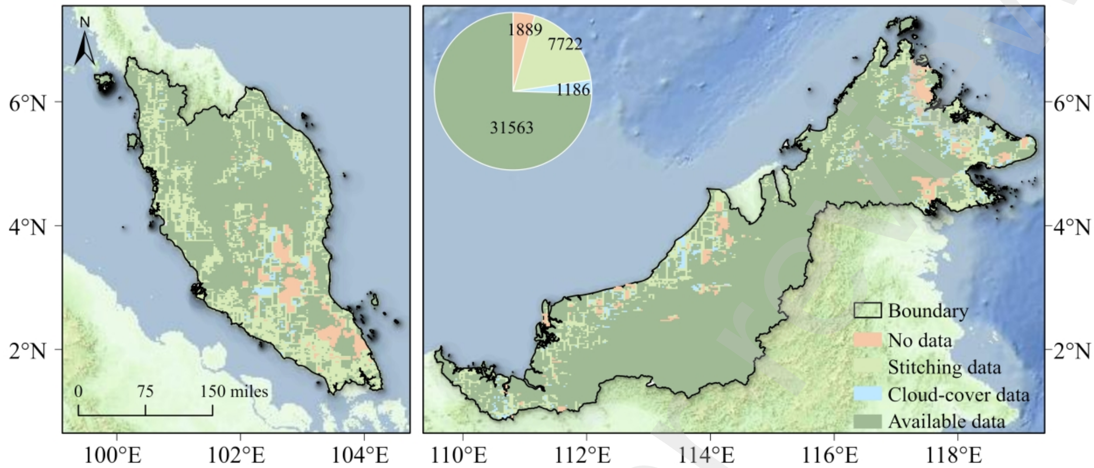

Fig. 2. Quality statistics and spatial distribution of satellite images across Malaysia.

2.2.2 Comparison and verification data

The comparison and verification data used in this study fall into two categories: (1) oil palm area and spatial distribution data, and (2) visual interpretation data for planting density and single tree of oil palm. (1) Oil palm area and spatial distribution data including the area compiled by the Food and Agriculture Organization of the United Nations (FAO), the 2018 oil palm distribution data generated by Xu et al. (2020) using PALSAR and MODIS data with the random forest method, the oil palm distribution data generated by Descals et al. (2021) using Sentinel data with deep learning methods, and the global oil palm data obtained by Meijaard et al. (2020) through various data integration. (2) We also collected global forest density data from Crowther et al. (2015), which can be compared with our forest density data. At the same time, distribution area and single tree position data of oil palm in 52 accuracy assessment areas were obtained through visual interpretation.

2.3 Spatial distribution and single tree extraction method of oil palm (STEM-OP)

2.3.1 Spatial distribution extraction of oil palm

(i) Gradient difference window detection

Oil palms present a series of unique regular features, which not only cover the spatial structure of the oil palm plantations, but also include the canopy features of single oil palm. The clustered and scattered leaves of oil palms in high-resolution remote sensing images show a radiating texture from the center to the surroundings, which provides unique visual features for information extraction (Barcelos et al., 2015; Tomlinson et al., 2006). In addition, oil palms artificial planting have also resulted in a relatively regular row pattern (Fig. 3).

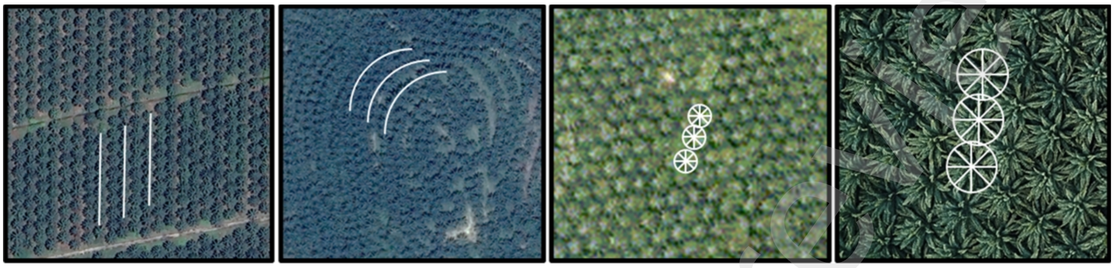

Fig. 3. Regular features of oil palm distribution and canopy.

This study transforms the regular features of oil palm plantations into morphological features. During this processing, a gradient difference window detection (GDWD) based on gradient difference of pixel brightness was constructed (Fig. 4). Specifically, a detection window was first defined, and for the pixels within the window, the brightness difference between the original pixel and the adjacent pixel was calculated in four directions from top to bottom, left to right, until reaching the window edge. Then, all the gradient differences of the window were compared with a specified gradient difference, and the morphological features of oil palm were characterized by setting the segmentation threshold. The feature quantity (E) is expressed by the formula:

$$f(x, y)_{up} = \sum_{i=0}^{d} 0 \quad ([n(x, y) - n(x, y + i)] > C) \tag{1}$$

$$f(x, y)_{down} = \sum_{i=0}^{d} 0 \quad ([n(x, y) - n(x, y - i)] > C) \tag{2}$$

$$f(x, y)_{right} = \sum_{i=0}^{d} 0 \quad ([n(x, y) - n(x + i, y)] > C) \tag{3}$$

$$f(x, y)_{left} = \sum_{i=0}^{d} 0 \quad ([n(x, y) - n(x - i, y)] > C) \tag{4}$$

$$E = \sum x,y \quad f(x, y)_{up} + f(x, y)_{down} + f(x, y)_{right} + f(x, y)_{left} \tag{5}$$

where, x and y denote the row and column coordinates of each grid point within the detection window, respectively; $f(x, y)_{up}$ represents the gradient difference; $n(x, y)$ is the brightness value at the grid point (x, y); The term $[n(x, y) - n(x, y+i)]$ indicates the difference in brightness values between the original grid point and the grid point after moving (i) coordinates in a certain direction; C is the specified gradient difference (threshold) and d is the detection window size. Simultaneously considering the gradient difference variations in four directions provides better feature identification for oil palm. Furthermore, the regular arrangement of clustered leaves and the orderly planting distribution of oil palm show significant differences in gradient response compared to other forests. Consequently, GDWD can capture the local changes of oil palm within the window area. This enables GDWD can more accurately capture the features of oil palm plantations.

(ii) Grayscale morphology

Grayscale morphology is an extension of binary morphology, capable of performing nonlinear processing on grayscale values within images (Serra, 2011; Shi et al., 2020; Wang et al., 2008). Typically, oil palm plantations exhibit relatively regular arrangement patterns, and often contain gaps between these arrangements within a window. This study further used two basic operations of grayscale morphology: erosion and dilation, along with four additional operations: opening, closing and top-hat transform. The purpose of erosion is to reduce high-value regions to minimize potential noise. The purpose of dilation is to enlarge high-value regions to compensate for any brightness gradient features that may have been lost during the erosion step (Zhang et al., 2023).

The input image was defined as $f(s, t)$, and the structuring element was defined as $b(x, y)$, where D_f and D_b represent the domains of the input image and the structuring element, respectively. The formula of dilation, erosion, opening, closing and top-hat transform are as follows:

$$(f \oplus b)(s, t) = max\{f(s - x, t - y) + \frac{b(x, y)}{(s - x, t - y)}\} \in D_f \mid (x, y) \in D_b \quad (6)$$

$$(f \Theta b)(s, t) = min\{f(s - x, t - y) - \frac{b(x, y)}{(s - x, t - y)}\} \in D_f \quad (7)$$

where, $f \oplus b$ represents the grayscale dilation and $f \Theta b$ represents the grayscale erosion. The result of grayscale dilation is brighter than the input image, enhancing the edges of bright areas in the original image, eliminating dark details with structuring element, and reducing most other dark details to some extent. The effect of grayscale erosion is exactly the opposite of dilation, which makes the output image darker than the input image. If the bright details in the input image are smaller than the structuring element, the brightness of the image will decrease.

$$f \circ b = (f \Theta b) \oplus b \quad (8)$$

$$f \cdot b = (f \oplus b) \Theta b \quad (9)$$

$$f' = f - (f \circ b) \quad (10)$$

Opening operation and closing operation are defined as $f \circ b$ and $f \cdot b$, respectively. Opening (/closing) operation is typically used to remove small bright (/dark) spots while preserving the grayscale levels. The top-hat transform is defined as f'.

(iii) OTSU

The OTSU method is an adaptive threshold determination method, also known as the maximum between-class variance method (Garcia-Murillo et al., 2020; Qiu et al., 2020; Shi et al., 2019). Applying the OTSU method to the gradient difference image adaptively finds the optimal segmentation threshold, thereby separating oil palm from other land cover types (Liu et al., 2021; McFeeters, 1996; Mohamed et al., 2022). When the between-class variance of two data sets is maximized, the misclassification error between the two groups of image data is minimized (Jing et al., 2012; Wang et al., 2008) (Fig. 4). By applying the OTSU method to the morphologically processed image, the optimal segmentation threshold can adaptively separate the oil palm plantation from other land cover type.

The equation used to calculate the between-class variance is:

$$\mu = \omega_0 * \mu_0 + \omega_1 * \mu_1 \quad (11)$$

$$g = \omega_0(\mu_0 - \mu)^2 + \omega_1(\mu_1 - \mu)^2 \quad (12)$$

where, ω_0 and ω_1 are the proportions of background and target pixels in one image, respectively; μ_0 and μ_1 denote the mean gray levels of the background and target pixels, respectively; μ represents the mean gray level of the image; The between-class variance is indicated by g. The optimal segmentation threshold between the background and target pixels was obtained when g reached its maximum value.

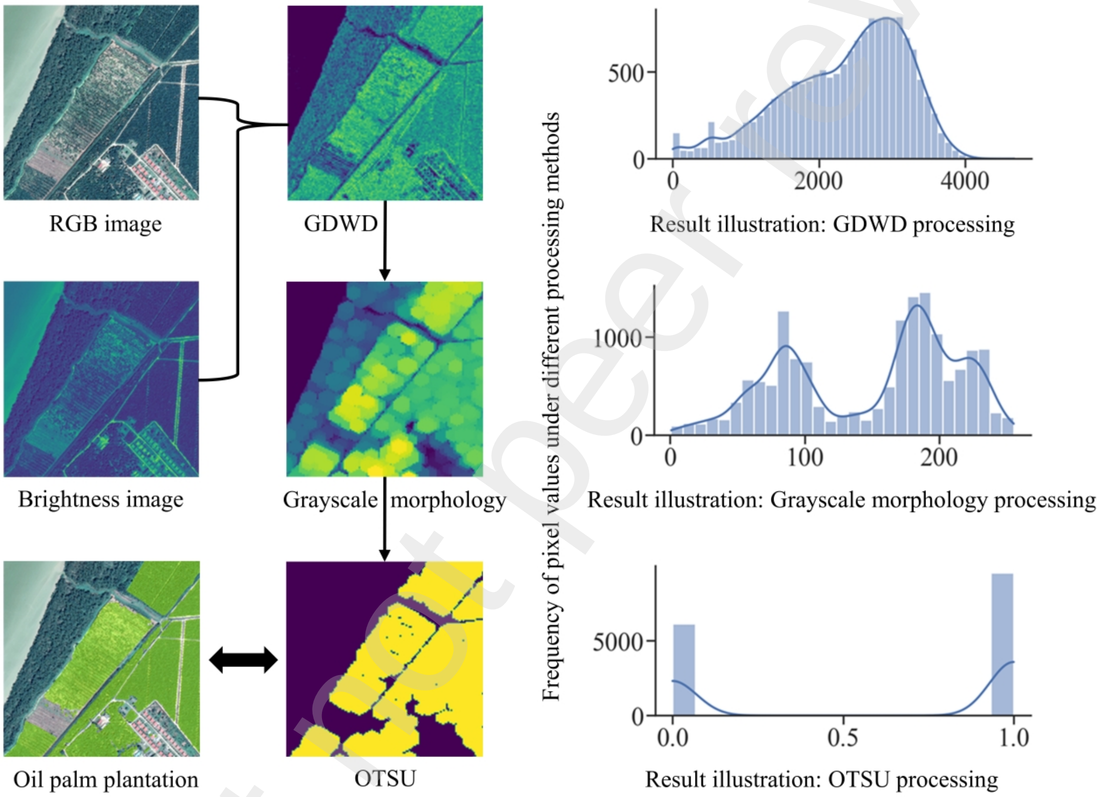

Fig. 4. Extraction process of oil palm plantations by gradient difference window detection (GDWD), grayscale morphology and OTSU.

(iv) Parameter adjustment

In GDWD, the specified gradient difference (threshold) C and the window size d are two critical parameters, which can be conducted to ensure optimal algorithm performance in oil palm extraction. A controlled variable adjustment and experimental approach were used to determine the optimal values of C and d, so as to maximize model accuracy under their mutual influences. In this study, C varied incrementally from 5 to 25 and *d* varied incrementally from 10 to 150. Fig. 5 shows the trend in identification accuracy of oil palm spatial distribution by GDWD under various parameter combinations. It can be observed that with the increase of *C* and *d* values, the identification accuracy initially increased and then decreased. When *C* was 15, the accuracy remained relatively stable across different window sizes. For parameter *d*, an optimal accuracy can be achieved when *d* reached 75.

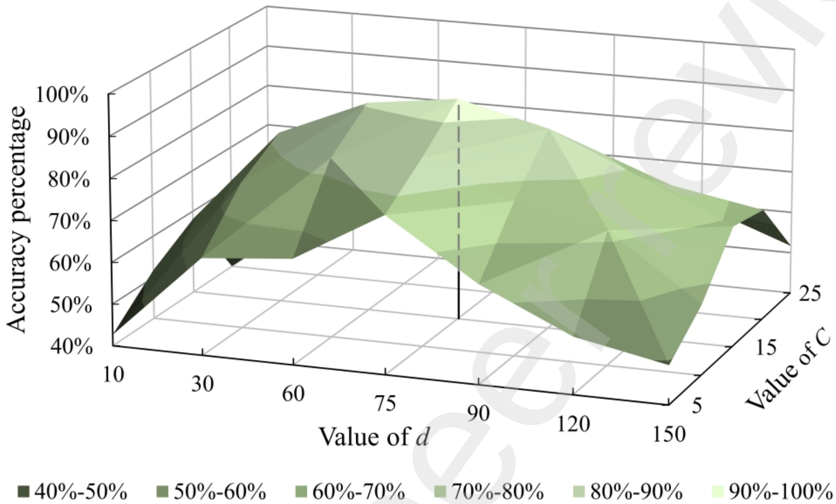

Fig. 5. Relationships between GDWD parameter changes and accuracy.

2.3.2 Information extraction of single oil palm

(i) Double ring sliding detection

The leaves of the oil palm emerge in clusters from the base of the trunk, fanning out to form a distinctive and orderly morphological structure. Through artificial pre-learning, we develop a new algorithm for extracting single tree information based on the unique canopy feature of oil palm, namely the double ring sliding detection (DRSD). In DRSD, the entire canopy of a single tree is considered the outer ring, while the distance halfway to the periphery of the canopy forms the inner ring. For the two rings, the ratio of high leaf values to low shadow values is calculated to capture the distribution of leaf clusters and shadows within a single tree canopy, respectively. The leaves within the inner ring are more densely clustered and shadows are fewer than the outer ring (Fig. 6). The DRSD is an unsupervised information extraction method that incorporates prior knowledge and does not require sample data or training.

The DRSD uses a sliding window to compute the proportion of high values within the outer and inner rings for all grid cells and the differences between them, so as to represent the canopy texture of a single tree and generate grid data with single tree features. The proportion of grid values for the double rings in a sliding window is expressed as follows:

$$f(x\text{: } x + k,\ y + k)_{small} = \frac{\sum_{i=1}^{s} \sum_{j=1}^{s} n(x + i,\ y + j)}{s^2} \tag{13}$$

$$f(x\text{: } x + k,\ y + k)_{big} = \frac{\sum_{i=1}^{b} \sum_{j=1}^{b} n(x - \frac{b - s}{2} + i,\ y - \frac{b - s}{2} + j)}{b^2} \tag{14}$$

where, $f(x\text{: } x + k,\ y + k)_{small}$ represents the detection value from the upper right corner $f(x, y)$ to the lower left corner $f(x+k, y+k)$ of a small-scale window; $n(x, y)$ is the grid value at point (x, y); k denotes the sliding step; s and b are the window size of inner ring and outer ring, respectively. Based on the growth features of oil palms and multiple measurements from satellite images, here we set half the radius for the sliding step, and for the original input image with a spatial resolution of 0.15 m, the parameters were $b = 40$, $s = 20$, and $k = 10$.

The specific processing of DRSD including morphological preprocessing and double ring detection. (1) Morphological preprocessing: In the identification processing of single oil palm, top-hat transformation was first applied to highlight areas with relatively high brightness, emphasizing leaf value areas. Then the Otsu's method was used to segment the image into two categories: high-value areas marked as 1 and low-value areas as 0. This binarized result more clearly presented the leaf distribution within the oil palm canopy. (2) Double ring detection: The ratio of high leaf values to low shadow values within the inner and outer rings of the sliding window was calculated, reflecting the relative density of leaves within the window. A window with single tree features showed a higher ratio in the inner ring and a lower ratio in the outer ring, then a coarse grid with single tree information was obtained (Fig. 6).

(ii) Adaptive iterative erosion algorithm

Due to the typically dense arrangement of trees, the coarse grid with single tree obtained from the DRSD cannot completely divide into single trees, and there are narrow and fragmented noise points between single trees (Fig. 6). This study used erosion algorithm to separate connected grids into individual patches. During the process of iterative erosion (Zhang et al., 2023), the smallest circular kernel of 3 was used in combination with the oil palm canopy features. Although the erosion strength is well-controlled using the smallest kernel value of 3, it is hard to precisely control of the erosion state for each canopy patch since continue eroding will lead to excessive erosion or disappearance of a coarse grid with single tree (Fig. 6).

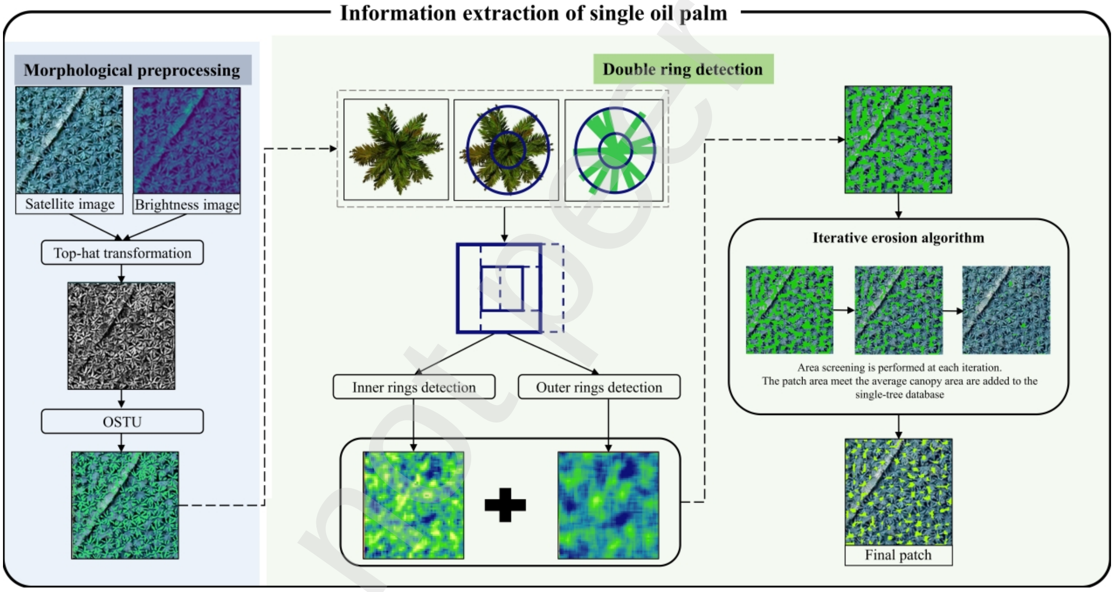

Fig. 6. Extraction process of single oil palm by Double ring sliding detection (DRSD) and adaptive iterative erosion algorithm (AIE).

To address this critical issue, adaptive iterative erosion algorithm (AIE) was constructed to adaptively control the number of iterations. After each erosion, canopy patches meeting the area threshold were added to the single tree data set, while remaining grids continue to the next erosion. The single oil palm canopy area used by DRSD was also used to define the area threshold. This ensures that smaller canopy patches are immediately separated when they meet the area requirement, while larger patches require continued iterative erosion. Finally, the separated single canopy patches were converted from grid to centroid points so as to obtain specific information of single oil palm.

2.4 Accuracy Verification

2.4.1 Accuracy Evaluation Method

In this study, various metrics from the confusion matrix are used to evaluate the performance of the spatial distribution and single tree information of oil palm. These include Recall rate, Precision rate, and F1-score.

$$R = TP/(TP + FN) \tag{15}$$

$$P = TP/(TP + FP) \tag{16}$$

$$\text{F1 - score} = 2R \times P/(P + R) \tag{17}$$

where, TP (True Positive) represents the correctly identified oil palm, while FN (False Negative) represents the oil palm that is mistakenly identified as other land cover type; FP (False Positive) represents the no oil palm that is mistakenly identified as oil palm. A high Recall rate indicates that the algorithm can more comprehensively identify oil palms. A high Precision rate means that the algorithm is more accurate in identifying oil palms, indicating fewer false errors. A high F1-score indicates that the algorithm has achieved a good balance between Recall rate and Precision rate, and has better overall performance.

2.4.2 Evaluation method for single tree information extraction accuracy

Count accuracy. To clearly present the identification results of number of single oil palm, this study selected a 200 m × 200 m area at the center of each of the 52 accuracy assessment areas (1km × 1km) as the accuracy assessment zones of single tree number. The single tree identification results were compared with those obtained through visual interpretation to evaluate the count accuracy of single tree identification.

Position accuracy. This study utilized nearest neighbor analysis to evaluate the spatial proximity between points identified through visual interpretation and those identified by STEM-OP. For each visually interpreted point, the closest point identified by STEM-OP was found, i.e., the point with the shortest distance. This nearest neighbor distance represents the positional deviation. Also, *R*, *P*, and F1-score were used to quantify the accuracy of single oil palm position. During this process, buffer zones were established around the visually interpreted oil palms. Buffer zones of varying distances served as the effective identification distance (Fig. 7).

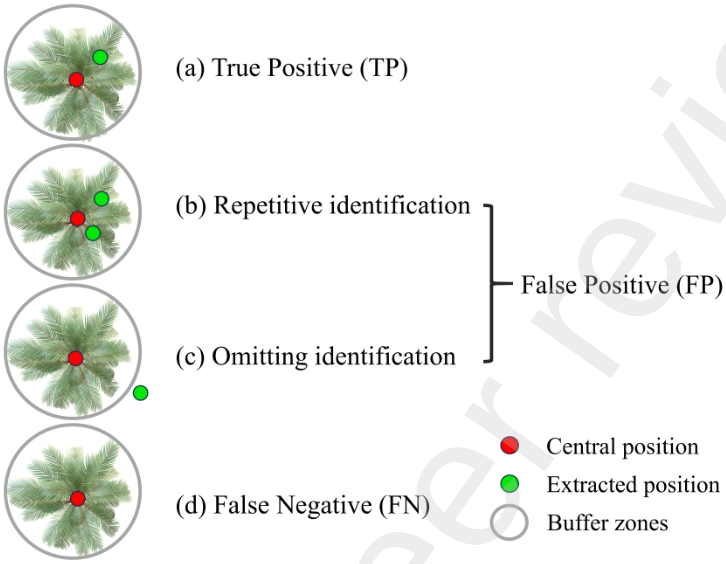

Fig. 7. Accuracy assessment of single tree position.

3 Results

3.1 Accuracy verification

3.1.1 Verification of spatial distribution accuracy

In the 52 accuracy assessment areas, this study compared the spatial distribution of oil palm using our algorithm with visual interpretation result. The graphic morphological algorithm GDWD yielded superior results in Recall rate, Precision rate, and F1-score. Specifically, the overall Recall rate reached 94.41%, the Precision rate was 91.08%, and the F1-score was 92.71%. These findings align consistently with visual interpretation, underscoring the efficacy of the algorithm (Table A.1).

Furthermore, in order to scrutinize the results of the GDWD, an examination was conducted by calculating the area percentage of the oil palm extracted by GDWD to the total area of the accuracy assessment areas. The coefficient of determination (R^2) for the area percentage was 0.91, indicating a notably high correlation between the two variables. Particularly in the accuracy assessment areas of 4, 8, 20, 30, 31, 33, and 42, the extracted oil palm area closely matched the visually interpreted area, with differences within 1% (Fig. 8). At the same time, due to the inconsistent quality of source data, the wide distribution of oil palm, the great differences in geographical environment and planting management in different regions, and the diverse types of surrounding land cover, some errors in extraction also occurred (Fig. A.1).

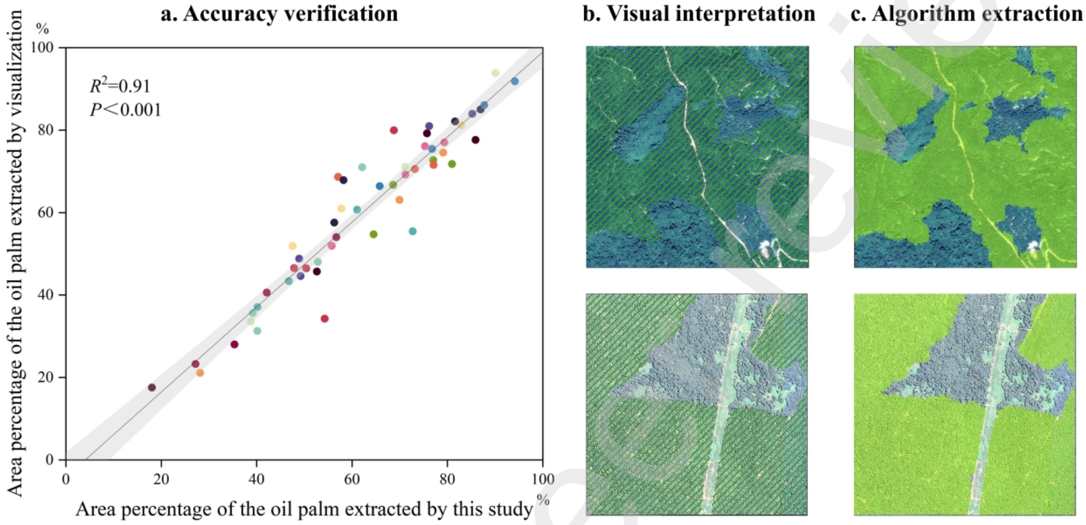

Fig. 8. Scatter plot of oil palm area accuracy.

3.1.2 Verification of single tree information extraction accuracy

To clearly present the extraction results of single tree information, the 200 m × 200 m central zones of accuracy assessment areas were selected. During this process, it was observed that 5 central zones of 8, 9, 18, 29, and 40 had no oil palm in the 52 accuracy assessment areas. Therefore, we used a total of 47 accuracy assessment zones to evaluate the accuracy of single oil palm. Here the identification results of single oil palm by STEM-OP were compared with those of visual interpretation. Across the 47 assessment zones, a total of 14,790 oil palms were extracted. Compared to the visually interpreted count of 15,746 trees, this study missed 956 trees, yielding an overall count accuracy of 93.93% (Table A.2). Overall, algorithms-extracted number and visual count of single oil palm maintain significant consistency too (Fig. 9).

The single tree position accuracy was evaluated by comparing the spatial proximity of visual interpretation points with algorithm-identified points. For each visual interpretation point, the nearest neighbor Euclidean distance to the point represents the deviation of position accuracy. The distance revealed that the deviation within the 47 assessment zones was consistently below 11.5 m. Among them, the distance deviation of 45% of these points was within the range of 3 m; and the distance deviation of 94% was within the range of 6 m (Fig. 9). Further detailed evaluation also revealed that at a distance deviation of 1-6 m, the three accuracy evaluation values (Recall rate, Precision rate, and F1-score) all increased rapidly. However, after exceeding 6 m, the three accuracy evaluation values increased, but relatively slowly (Table A.3). At the same time, we found that the quality of satellite source data mainly affected the accuracy of extracting single tree numbers and position information (Fig. A.2).

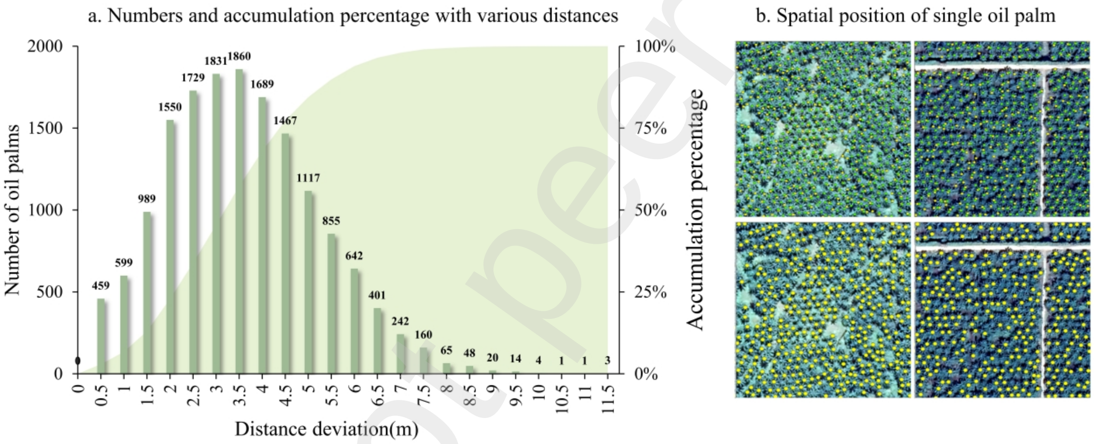

Fig. 9. Number of various distances and spatial position of single oil palm. Fig. 9a shows the number and accumulation percentage of single oil palm with various distances; 9b shows the spatial distribution of single oil palm. The two upper images show the comparison between algorithm-identified points (yellow) and visually interpreted points (green), while the tow lower images indicate the correctly identified points (bigger yellow points).

3.2 Analysis of oil palm distribution and planting density

3.2.1 Distribution pattern of oil palm in Malaysia

Oil palm was widely distributed in Malaysia, especially in West Malaysia (Fig. 10). The oil palm plantation in Malaysia is 5.252 million ha, accounting for 15.90% of the national total land area. The oil palm plantation in West Malaysia accounts for 24.58%

of its land area, while the proportion in East Malaysia is 12.55%. Notably, Sabah and Sarawak, located in West Malaysia, possess the largest oil palm plantations, reaching 112.63 and 135.97 hectares respectively. Combined, these two states contribute approximately half of the nation's total plantation area. Conversely, Perlis in East Malaysia exhibits the smallest oil palm plantation, comprising only 15,800 hectares, considerably less than other regions. Malacca and Penang also demonstrate relatively modest plantation areas. Additionally, Johor, Kedah, Kelantan, Pahang, Perak, and Terengganu boast plantation areas exceeding 200,000 hectares, signifying significant distribution of plantations within these states, while Negeri Sembilan and Selangor possessed plantation areas exceeding 100,000 hectares, comparatively smaller in area.

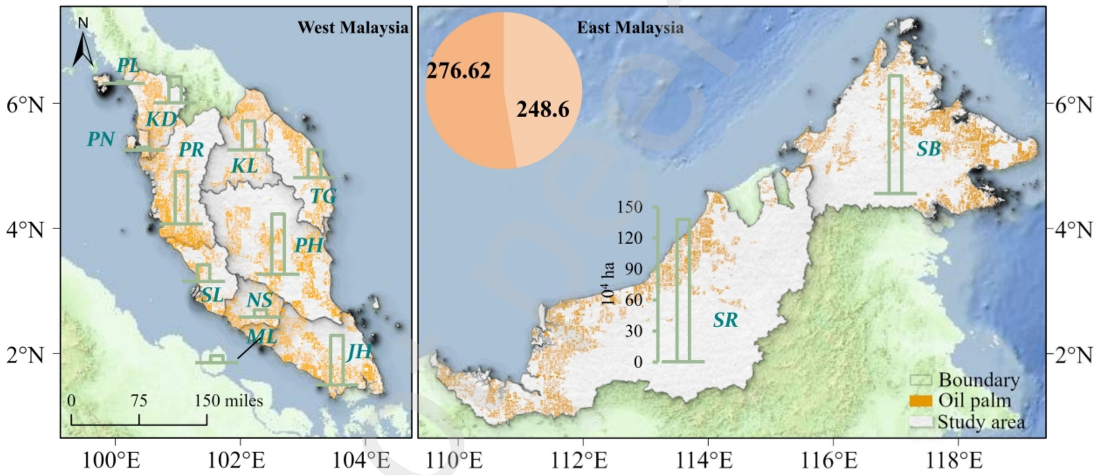

Fig. 10. Spatial distribution of oil palm in Malaysia in 2021 and 2022.

3.2.2 Planting density of oil palm

The study generated a nationwide planting density map of oil palm in Malaysia (Fig. 11). We extracted a total of approximately 572.6 million oil palms across Malaysia, with an average planting density of 109 palm per ha. Notably, West Malaysia accounted for the majority, with around 297.48 million oil palms, while East Malaysia hosted approximately 275.13 million. At the state level, oil palm in Sabah and Sarawak in East Malaysia exceeded 100 million. In West Malaysia, three states, Johor, Pahang, and Perak all had over 50 million palms and Pahang had the most count of 62.19 million. Conversely, Malacca, Perlis, and Penang exhibit relatively fewer oil palms.

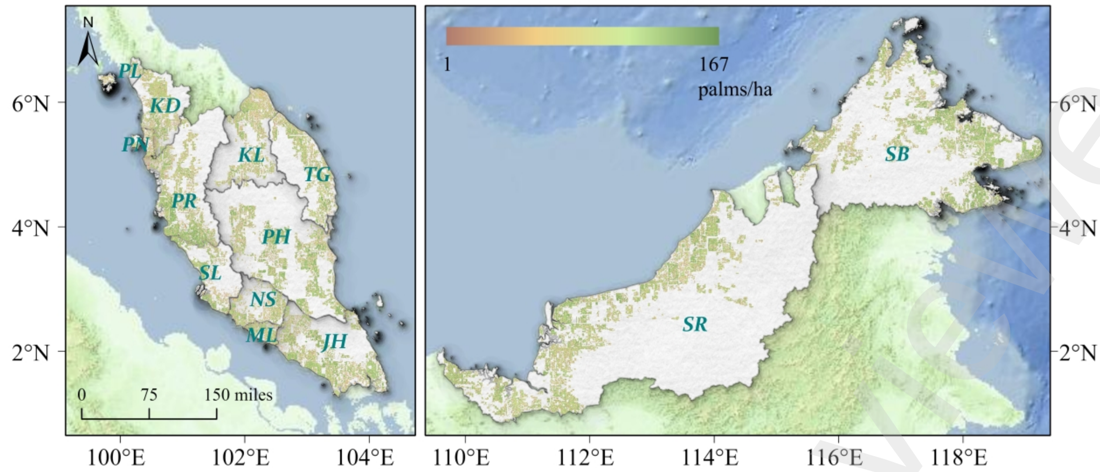

Fig. 11. Oil palm planting density map in Malaysia.

Regarding average planting density, variations exist among states, yet they predominantly fall within the range of 100 to 110 palms per hectare. Perak exhibited the highest planting density, reaching 111 palms per hectare, indicative of relative intense oil palm in the state. Conversely, Malacca and Penang showed relatively lower planting densities of 95 and 97 palms per hectare, respectively. These differences may stem from diverse factors such as land use practices, climatic conditions, or variations in planting techniques within these regions.

4 Discussion

4.1 Applicability of STEM-OP

This study proposes STEM-OP, a method that extracts both spatial distribution and single tree information of oil palm without the training samples and learning. Unlike machine learning or deep learning, STEM-OP thoroughly explores the visual features of oil palm in sub-meter satellite image. By detailed analysis of the morphological features of oil palm plantation and single tree canopy, STEM-OP were used to extract the spatial distribution and single tree information of oil palm. Compared to traditional methods or machine learning algorithms (Arevalo-Ramirez et al., 2021; Dechesne et al., 2017; Osco et al., 2021; Zheng and Wu, 2022), the STEM-OP not only does not require pre-training samples and other hyperspectral or radar data, but also keeps high accuracy (Dong et al., 2020; Ke et al., 2010; Somers and Asner, 2014) as well as consumes fewer computational resources and achieving higher computational efficiency. During the processing, the hardware conditions of this study included an Intel I7-12700 CPU with a frequency of 4400 MHz, 12 cores, 20 threads, and 16 GB of memory. For the 39.90 TB of 39,285 high-resolution satellite images, the total processing time of STEM-OP was 838.7 hours (Table A.4). Therefore, STEM-OP provides a powerful tool for large-scale, high-efficiency and finer mapping of oil palm, with low computational resource requirements and strong applicability value. Large companies and private plantations for planting oil palms will be interested in the cost-effectiveness of freely available RGB satellite image and methods versus using expensive drones and commercial satellites to extract oil palms. Additionally, STEM-OP is not only applicable to oil palm but also offers new ideas and technical means for extracting single tree information of other tree species, especially for oil palm-like trees (Zhang et al., 2023). However, STEM-OP is directly dependent on the quality of the source image. Despite utilizing sub-meter resolution satellite data, uncertainties arise due to the varying sensors, acquisition times, cloud interference, and particularly the image mosaicking.

4.2 Comparison of the accuracy of the extraction results

This study indicates that the oil palm plantation in Malaysia is 5.25 million hectares. Descals et al. (2021) reported an area of 5.45 million hectares in 2019, FAO reported 5.14 million hectares in 2021, and Meijaard et al. (2020) estimated the area to be 5.05 million hectares in 2017. Comparing our findings with other studies and publicly available data, all of which are relatively close to our results. However, Rodríguez et al. (2021) reported a significantly larger area of 6.17 million hectares in 2019, and Xu et al. (2020) reported 5.99 million hectares in 2018 (Table A.5). These two larger area data may be due to their Sentinel data with a 10-meter spatial resolution, which include roads and drainage ditches, thus inflating the plantation area.

Spatially, Crowther et al. (2015) published the first global forest density data. Since this data does not distinguish oil palms from other trees, we used pure pixels with full coverage of oil palm in 52 accuracy assessment areas and finally selected 30 accuracy assessment areas for density accuracy assessment. Comparing our results with the global forest density data, we found that the discrepancy between the two results could be attributed to the lower resolution (1 km) of the global forest density data, which including natural forests, or mixed tree-shrub composite within mixed pixels. Additionally, previous studies reported Malaysia's oil palm density as 87.5 palms per hectare and 130-140 palms per hectare (Asari et al., 2017; Corley and Tinker, 2008; Latif et al., 2003; Rodríguez et al., 2021). Our study found an average density of 109 palms per hectare, falling within the range of current study data. However, STEM-OP is constructed by pre-learning the morphological structure and canopy features of oil palm, which are directly limited by image quality such as cloud and clarity. In addition, the complexity of the surrounding environment around oil palm plantations and the small young forests will also interfere with the accuracy to a certain extent. Therefore, the accuracy of this study will be further improved through preprocessing of satellite images, such as cloud removal and super-resolution in the future.

This study confirmed the great potential of RGB satellite images in extracting the single-tree information of a certain tree species over large spatial scales. Through the example of the Malaysia's oil palm, we demonstrate how to use remote sensing data to conduct large-scale and detailed studies on tropical crops and forests. In addition, the method can be easily applied to other forests or other regions, thus contributing to the refined management of the global ecosystems and the response to global change.

5 Conclusion

This study focuses on extracting the spatial distribution and single tree information of oil palms on a national scale by thoroughly exploiting the morphological features of oil palm in sub-meter RGB satellite images. With incorporated pre-learned features, we developed a novel, sample-free method STEM-OP to extract the spatial distribution and single tree information of oil palm. For extracting spatial distribution, we developed a graphic morphological algorithm of GDWD. For extracting single tree information, we proposed DRSD and AIE algorithms.

The results showed that the STEM-OP had higher accuracy and the extracted spatial distribution achieved an overall Recall rate of 94.41%, Precision rate of 91.08%, and F1-score of 92.71%. Due to the quality limitations in satellite images, the specific accuracy varied among the 52 accuracy assessment areas, but it exceeded 90% in most of the accuracy assesssment areas. The overall counts accuracy of extracted oil palm reached 93.93% and 77% of position accuracy of single oil palm were within a 6-m deviation. For the entire country, the total area of oil palm plantations was 5.25 million hectares, accounting for 15.90% of Malaysia's total land area in 2011 and 2022. The overall counts were 572.6 million palms, with an average planting density of 109 palms per hectare. This study reveals the current state of oil palm plantation in Malaysia and provides direct methods and data support for refined assessing oil palm resources, monitoring impacts on biodiversity, and promoting the sustainable development of the oil palm industry.

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Sen Zhang: Writing – original draft, Methodology, Validation, Data curation, Formal analysis. **Nan Li**: Writing – original draft, Visualization, Software, Conceptualization. **Yaoping Cui**: Supervision, Methodology, Conceptualization, Writing – review & editing. **Jinwei Dong**: Supervision, Conceptualization, Writing – review & editing. **Le Yu**: Writing – review & editing; **Lishan Ran**: Writing – review & editing; **Ziyue Chen**: Writing – review & editing; **Jiqiang Niu**: Writing – review & editing; **Wei Yan**: Writing – review & editing; **Kasturi Devi Kanniah**: Investigation, Writing – review & editing; **Xiangming Xiao**: Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data used are available as open access. The current research results including single oil palm information data in Malaysia are freely available at https://doi.org/10.5281/zenodo.12684445.

Appendix A. Supplementary data

Supplementary data for this article are available online.

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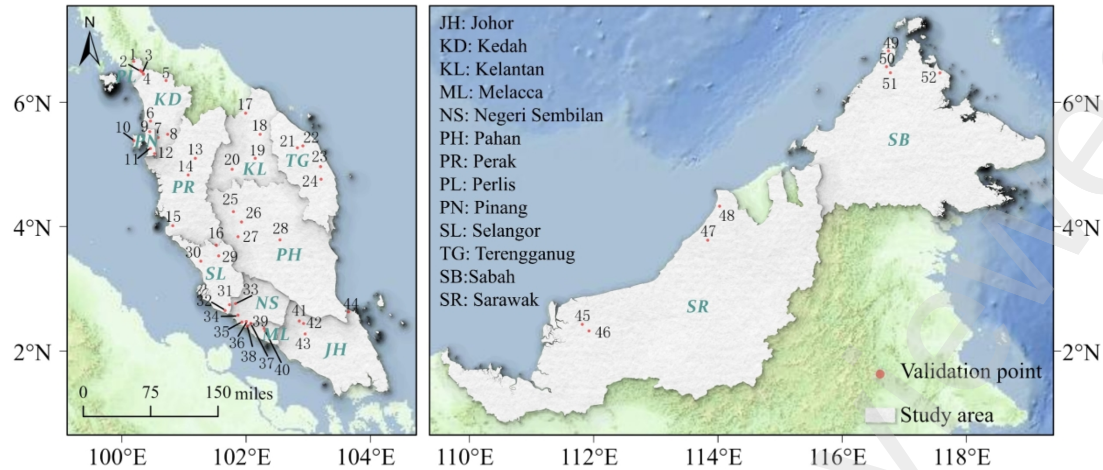

Fig. 1. Study area and the 52 accuracy assessment areas.

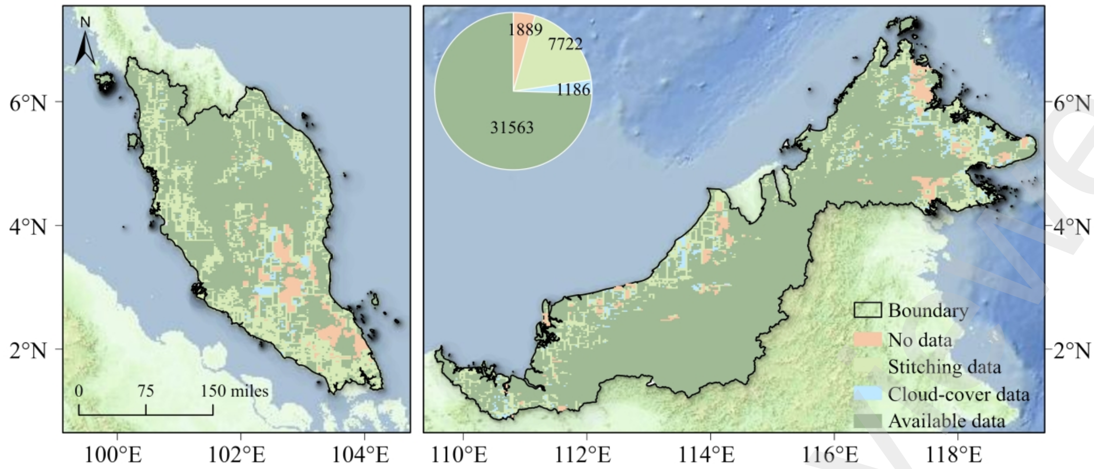

Fig. 2. Quality statistics and spatial distribution of satellite images across Malaysia.

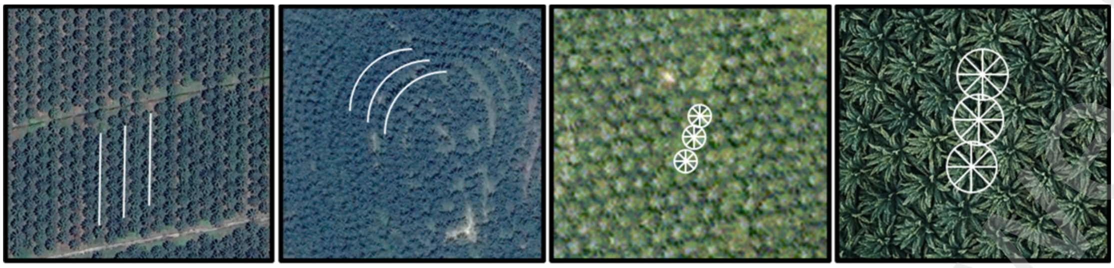

Fig. 3. Regular features of oil palm distribution and canopy.

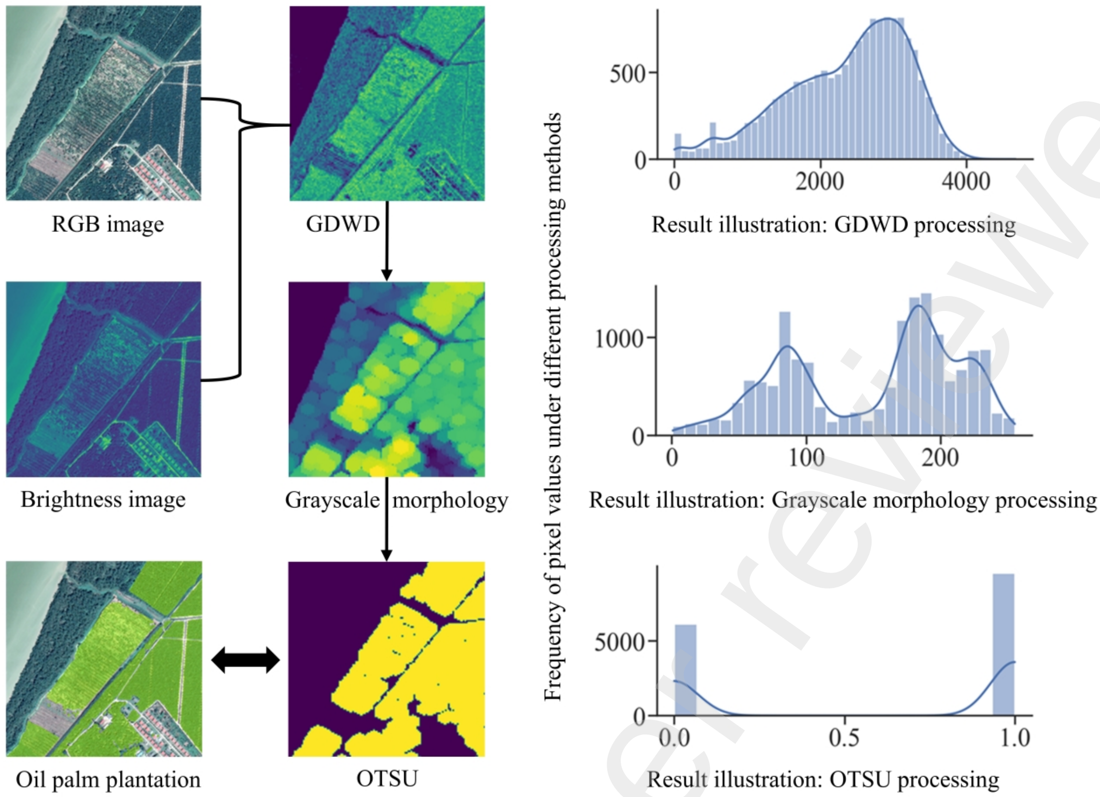

Fig. 4. Extraction process of oil palm plantations by gradient difference window detection (GDWD), grayscale morphology and OTSU.

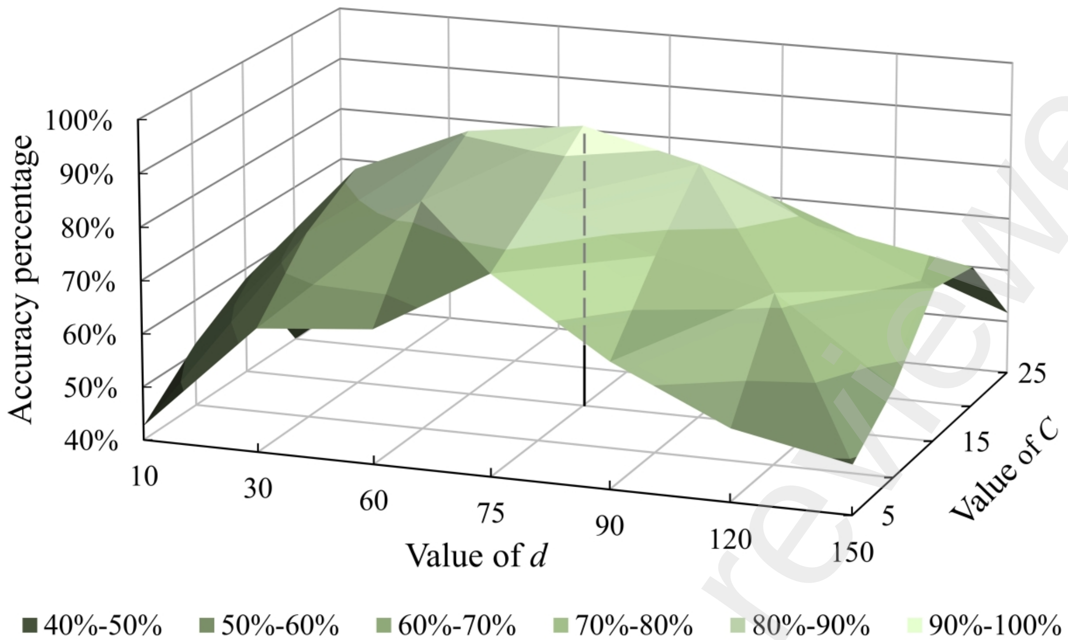

Fig. 5. Relationships between GDWD parameter changes and accuracy.

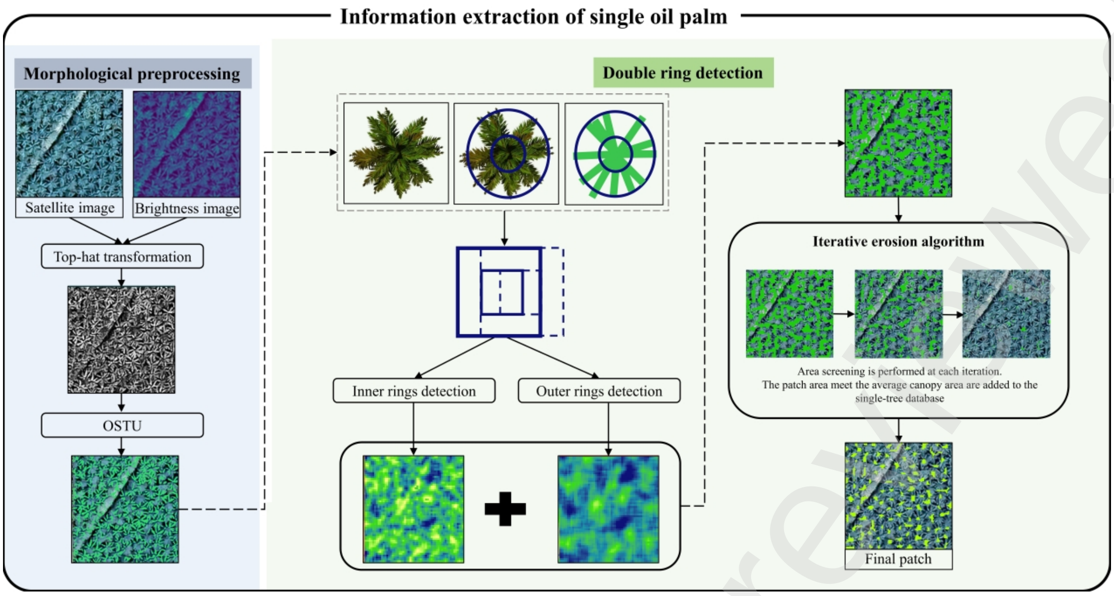

Fig. 6. Extraction process of single oil palm by Double ring sliding detection (DRSD).

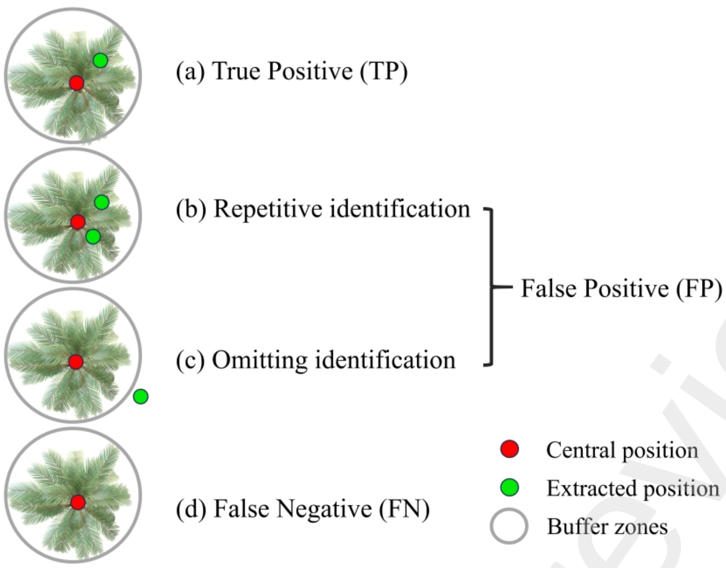

Fig. 7. Accuracy assessment of single tree position.

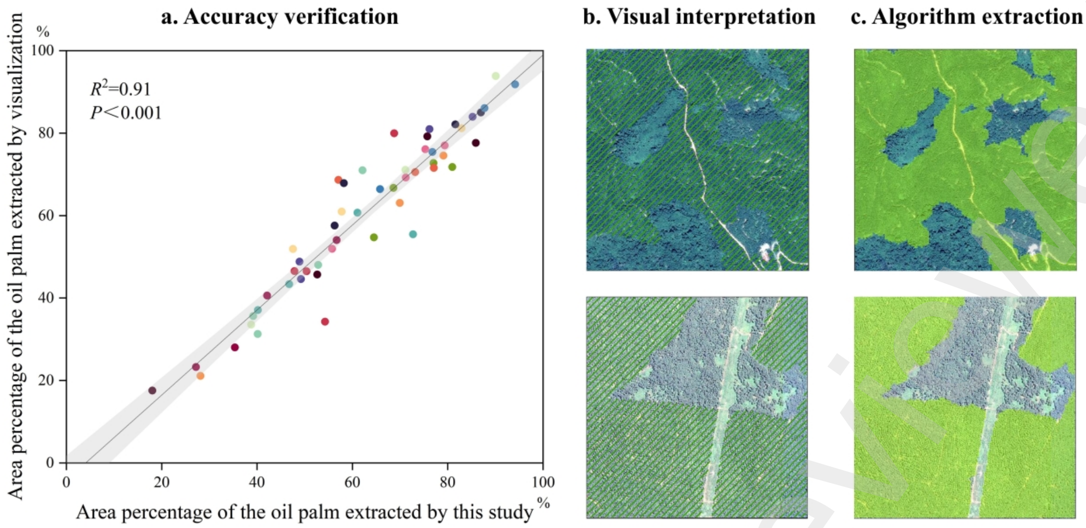

Fig. 8. Scatter plot of oil palm area accuracy.

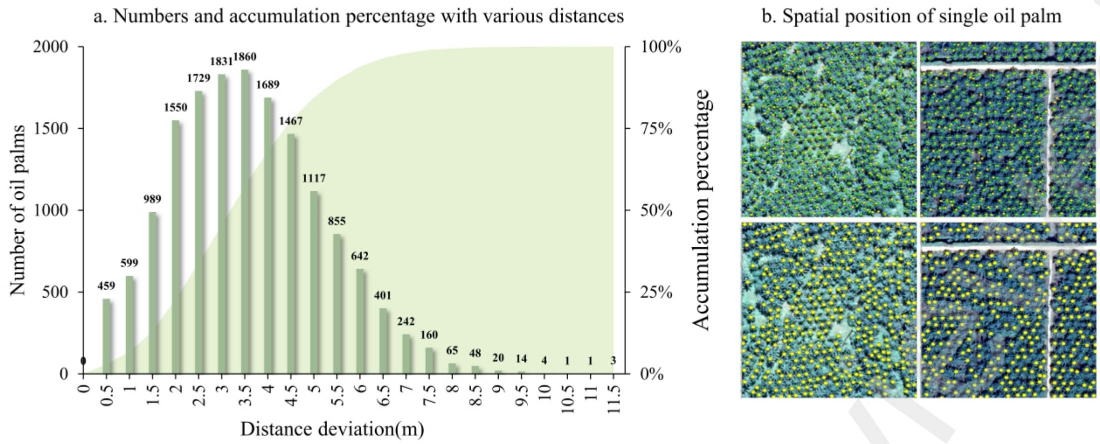

Fig. 9. Number of various distances and spatial position of single oil palm. Fig. 9a shows the number and accumulation percentage of single oil palm with various distances; 9b shows the spatial distribution of single oil palm. The two upper images show the comparison between algorithm-identified points (yellow) and visually interpreted points (green), while the tow lower images indicate the correctly identified points (bigger yellow points).

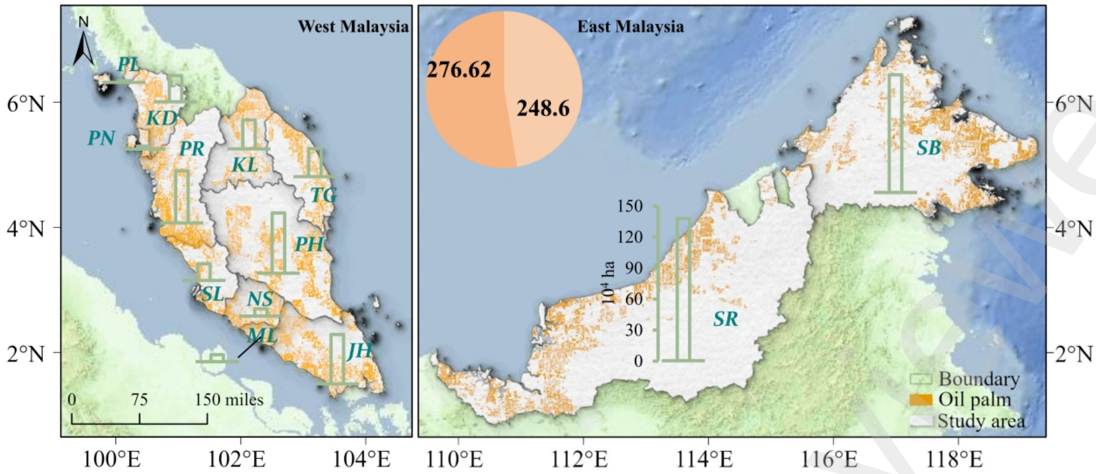

Fig. 10. Spatial distribution of oil palm in Malaysia in 2021 and 2022.

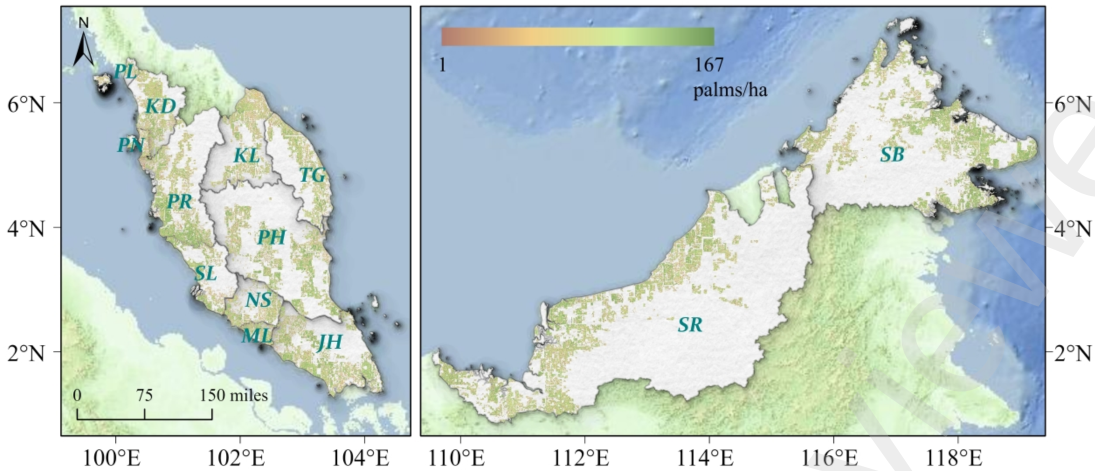

Fig. 11. Oil palm planting density map in Malaysia.

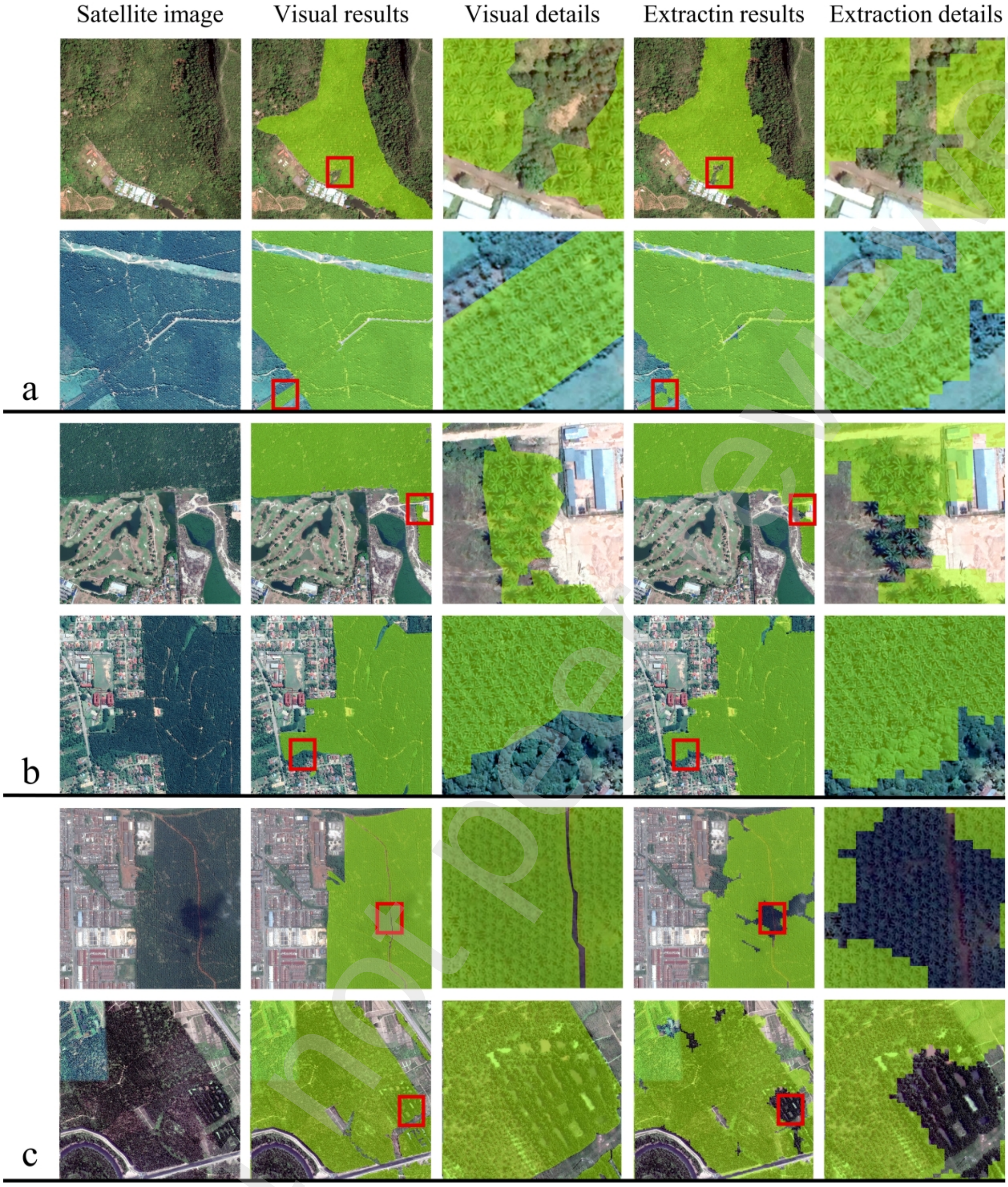

Fig. A.1. Details **of spatial distribution and three categories of extraction errors.** The geometric characteristics of the oil palm plantations and the quality of satellite images are the main factors affecting the oil palm extraction accuracy. Fig. A.1a shows the extraction results under edge geometric anomalies, which implies that the geometric features of these regions are relatively complex and sensitive to algorithm parameters. Fig. A.1b shows the extracted results under large spectra differences among various land cover types or other vegetation types. Large spectra differences and mixed spectra will directly affect the gradient difference of window detection. Fig. A1c shows the extraction results under the interference of cloud shadows and the splicing of various images, which directly leads to abnormal instability of spectral features in some areas within an image.

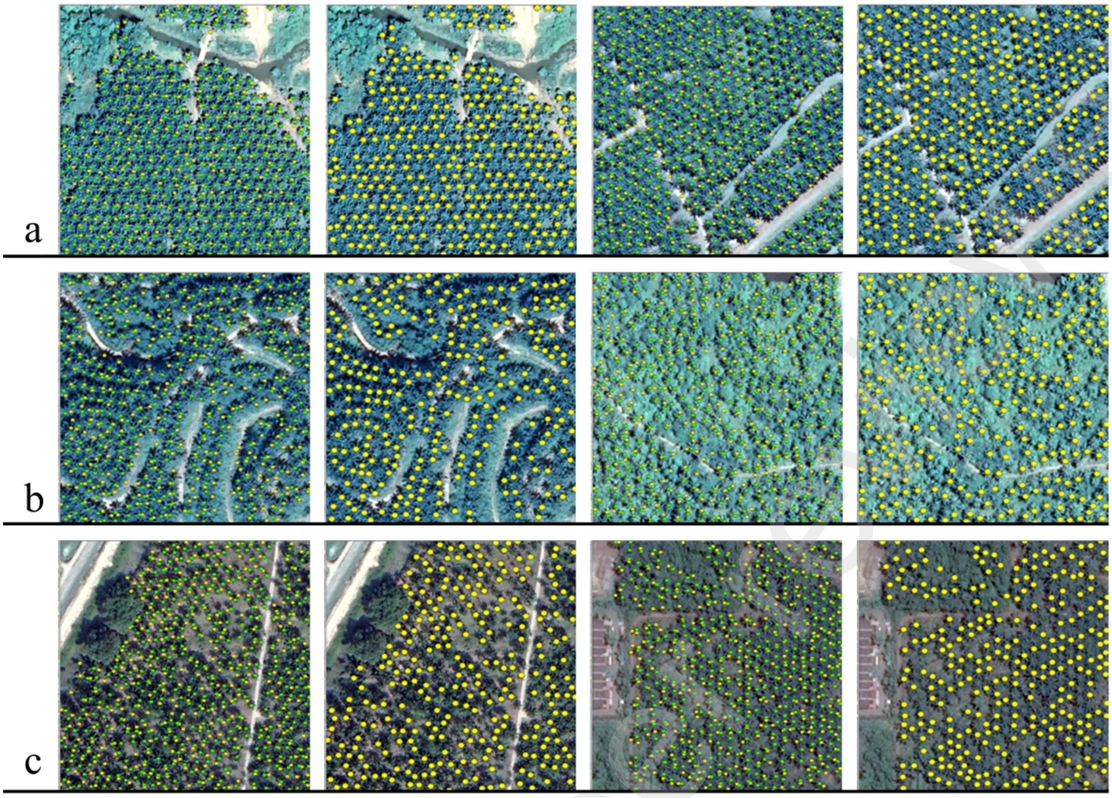

Fig. A.2. Spatial details of single oil palm and three categories of extraction errors. The left image shows a comparison between algorithm-identified points (yellow) and visually interpreted points (green), while the right image indicates the correctly identified points (bigger yellow points). Fig. A.2a shows the single oil palm in flat areas, where oil palms exhibit a consistent growth state. Fig. A.2b is the single oil palm in complex terrain area. Mountainous terrain results in uneven distribution of oil palms, increasing the occurrence of overlapping tree canopies. Fig. A.2c shows the single oil palm under interferences of shadows or coarse-resolution satellite image. Tree shadows obscure the edges of the oil palm canopies, increasing the difficulty of identification. Additionally, STEM-OP heavily relies on the spatial resolution of the original satellite image, the clarity of the original images thus directly impacts the algorithm's accuracy.